Econometrics 2 — Midterm assessment

Applied Project: Labor Market and Origin

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Presentation This project aims to measure the effect of various individual variables on labor market position, studied here along an extensive margin (employed, unemployed, inactive) and an intensive margin (hourly wage). The main explanatory variable of interest is a categorical variable "origin" constructed from the nationalities of the individual's parents at birth.[1]

General Instructions The project is not based on a specific article and is exploratory in nature. It allows some leeway in the choice of estimated models, but in return requires always being precise and clear about: *(i)* the observations used, *(ii)* the variables included in the analysis, *(iii)* the estimation and inference method (computation of standard errors). Careful attention will be paid to the choice of models, the discussion of their assumptions, the parameters of interest considered, and the qualitative and quantitative interpretation of the resulting estimates.

Data The dataset contains $n = 10{,}000$ individuals observed over 6 quarters, between 2014 and 2016 (balanced panel) from the Labor Force Survey. For each observation (individual $\times$ quarter), the following variables are available:

- Demographic and individual socio-economic variables: age, sex, education, parents' socio-professional category, "origin", an indicator for being an immigrant ("immi"), an indicator for being the child of an immigrant ("desc"), self-reported health status, marital status, etc.
- Individual variables related to labor-market position: status (employed, unemployed, inactive), wage, hours worked, type of working hours, type of employment contract, etc.
- Variables related to the residential area: urban unit size category, urban or rural municipality, proportions of populations of different origins in the area, etc.

Appendix A presents the detailed variable descriptions.

1 Intensive Margin (Net Hourly Wage)

In this first part, we study the wages of employed individuals and examine the existence of potential wage inequalities due to origin.[2] The wage is only available in the first and last quarter: panel data with $T = 2$ for this variable.

The concept of wage inequality due to origin is similar to that of gender wage inequality. There is a close link with the econometric approach: we will speak of wage inequality due to origin when, *all else equal*, individuals' wages differ by origin. The underlying idea being that, if these wage differences are not justified (since we hold everything else equal), they can be interpreted as inequalities.[3] A crucial question, therefore, concerns which explanatory variables are included in the "all else equal."

[1] The "origin" variable takes nine values here: France, Northern Europe, Southern Europe, Eastern Europe, Maghreb, rest of Africa, Middle East, Southeast Asia (Cambodia, Laos, Vietnam), rest of the world.

[2] We ignore here the censored nature of the wage variable by restricting to individuals for whom the wage is observed (employed individuals).

[3] These explanations aim to clarify the issues of this project. They obviously do not exhaust the question of inequalities, which involves multiple dimensions: equality versus equity, the situation at a given point in time versus a dynamic analysis over an individual's lifetime, etc.

Question 1 We ignore the panel dimension for now by using only the last observed quarter for employed individuals. We consider a linear model with net hourly wage (variable "salhoraire") or the logarithm of net hourly wage (variable "logsalhoraire") as the dependent variable.

a) Estimate each of these two models for a chosen set of explanatory variables and justify your choices. Different specifications may be tried and presented. Attention should be paid to endogeneity problems and the distinction between correlation and causality.

b) Interpret quantitatively and qualitatively the estimated coefficients for the explanatory variable "origin," when the dependent variable is the net hourly wage and when it is the logarithm of the net hourly wage.

c) Through what channels does the effect of the "origin" variable manifest on wages? To address this question, one may examine how the estimated coefficients of the "origin" variable evolve depending on the explanatory variables included in the model: education, types of employment contracts, sectors of activity, etc.

Question 2 Education level is an important determinant of wages, and in this question, we wish to include it as a control variable.

a) What problem could arise from including this explanatory variable (which partly results from individual choices) in the models estimated in Question 1? What would be the effect, in particular, on the estimation of the coefficient of the explanatory variable "origin"?

b) Among the variables available in the dataset, propose variables that could be used as instruments for education level and discuss their validity. Pay attention to the required number of instruments.

c) Estimate the corresponding model in which the education level is instrumented by the variables determined in the previous question.

d) Interpret the results obtained in light of the research question. In particular, provide a quantitative interpretation of the estimated coefficient for the explanatory variable "origin."

Question 3 Under some assumptions, panel data can also resolve endogeneity problems. [4] Unlike the two previous questions, we use here the panel structure of the data for wage-related variables: we consider observations from the first and last quarter for employed individuals.

a) Estimate a model in which the exogeneity of residuals and individual fixed effects is assumed.[5]

b) Is it possible to include the "origin" variable in the previous model (Question 3.a)? Does this model allow potential endogeneity problems to be resolved?

c) We now allow individual effects to be correlated with the regressors. With the methods covered in class, is it possible in this framework to study the effect of the "origin" variable?

[4] For all questions on panels, keep in mind the structure of the course and the distinction between: the case where the unobserved individual terms, denoted α_i in the course, are assumed exogenous (the framework often called "random effects"); and the case where the individual terms are allowed to be correlated with the regressors (the "fixed effects" framework).

[5] You may use the specifications discussed in Question 1.

2 Extensive Margin (Employed vs. Unemployed)

The effect of origin on the labor market can also occur along an extensive margin: having or not having a job. In this section, we focus on active individuals (employed or unemployed) with the indicator of being unemployed as the dependent variable. Labor market status (employed, unemployed, inactive) is available each quarter (panel data with $T = 6$).

Question 4 The dependent variable is binary. We ignore this limitation for now and, in this question, consider a linear probability model. We take into account the panel structure of the data.

a) Estimate a panel model in which the exogeneity of residuals and individual fixed effects is assumed, and interpret the results.

b) Again, what problem arises if we wish to allow for correlation between individual effects and the regressors in order to study the effect of the "origin" variable?

Question 5 In this question, we set aside the effect of the "origin" regressor to focus on a dynamic model and study state dependence: we aim to estimate the effect of being unemployed at a period t on being unemployed at the next period $t + 1$ (one period corresponds to one quarter).

a) Discuss the credibility of the strict exogeneity and weak exogeneity assumptions in this context, and estimate such a dynamic model.

b) Interpret the results obtained.

Question 6 We now take into account the fact that the dependent variable is binary and consider a probit or logit model for the indicator of being unemployed.

a) Are the within and first-difference panel methods, which allow the individual fixed effect α_i to be eliminated, applicable here?

b) We restrict to cross-sectional data using the last quarter. Estimate a probit model and a logit model. Compare the estimated marginal effects for the explanatory variable "origin" between the two models and interpret the results.

c) We would like to control for education level, but this may raise an endogeneity problem. To simplify, from the variable "education," we construct the indicator of having a higher education degree (modality "1," see Appendix) and include it as an explanatory variable, together with "origin" and any relevant other covariates, depending on your chosen specification. Implement the Rivers-Vuong method studied in the tutorial sessions to estimate such a model in which having a higher education degree is not assumed to be exogenous. Comment on the results obtained.

A Appendix: Dataset Description

The dataset contains 60,000 observations and 81 variables. It is in Stata format (.dta).

Note: A number of variables already have clear labels and will therefore not be detailed below when their values present no particular difficulty (for example, the dataset contains a variable dep labeled "département" with values ranging from 01 to 95). Furthermore, the values "." for numeric variables and "" for "character" variables denote — following Stata convention — missing or non-applicable values, i.e., undefined by construction. These can, of course, be taken into account when selecting the observations used in the analyses.

As explained in the questions, you have relative freedom in choosing control variables and precise model specifications; it is probably not necessary to use all of these variables. On the contrary, the aim is to identify and select those that seem most relevant for addressing the research question in a convincing way, and to justify those choices. Defining new variables is perfectly acceptable — for instance, by grouping the values of certain variables, or by considering interactions, squared terms, etc. Similarly, it will likely be useful to restrict the sample to a population of interest, whichever seems most relevant for answering the question.

An observation is uniquely identified in the dataset by an individual × period pair, using the variable `id_individu` as the individual identifier (i.e. an individual is uniquely identified by a value of `id_individu`) and the variable `t`, ranging from 1 to 6 (individuals are followed for 6 quarters). This is thus panel data with $n = 10{,}000$ individuals and $T = 6$ time periods. The panel is balanced.

A household identifier is also available. Individuals within the same household share the same value of `id_menage`. Within a household, individuals are numbered by the variable `num_ind_par_menage`. This additional level of the data will not be used in the analysis. It may nonetheless be noted that `num_ind_par_menage` takes only two values, 1 and 2. This calls for a few words on the sample selection and dataset construction — i.e., how observations were selected. The individuals in the dataset are the household reference person and, where applicable, their spouse. These are also individuals who could be interviewed over six consecutive quarters of the survey: they did not change residence (the survey is conducted by randomly sampling dwellings whose occupants are interviewed, or at least sought to be interviewed), and they were reachable and responded to the survey.

Below is a detailed list of the variables in the dataset, excluding those whose Stata label and values are sufficiently self-explanatory.

- `rim`: household interview rank. Households are interviewed over six consecutive quarters in principle, but certain types of individuals are only interviewed twice, at the first and sixth quarter. *Note: it may be worth identifying which individuals are concerned [hint: look at their age], and it is entirely justified to exclude them from the analysis.*
- `typmen`: household type: 1 = single-person household, 2 = single-parent family, 3 = couple without children, 4 = couple with child(ren), 5 = multi-person household where all members are related to the household reference person, neither a couple nor a single-parent family, 6 = multi-person household where not all members are related to the household reference person, neither a couple nor a single-parent family, 9 = other complex households of more than one person.
- `region`: (using names prior to certain regional renamings) 1 = Île-de-France, 2 = Bretagne, 3 = Aquitaine–Limousin–Poitou-Charentes, 4 = PACA, 5 = Auvergne–Rhône-Alpes, 6 = Alsace–Lorraine Champagne-Ardenne, 7 = Midi-Pyrénées–Languedoc-Roussillon, 8 = Nord-Pas-de-Calais–Picardie, 9 = Normandie, 10 = Pays de la Loire, 11 = Centre–Val de Loire, 12 = Bourgogne–Franche-Comté, 13 = Corse, 14 = outside metropolitan France (overseas departments).
- `tuu`: urban unit size category of the dwelling (a dwelling is associated with a single household and the individuals composing it; since the dataset is restricted to individuals who did not change residence, as explained above): 0 = Rural municipality, 1 = Urban unit of fewer than 5,000 inhabitants, 2 = Urban unit of 5,000 to 9,999 inhabitants, 3 = Urban unit of 10,000 to 19,999 inhabitants, 4 = Urban unit of 20,000 to 49,999 inhabitants, 5 = Urban unit of 50,000 to 99,999 inhabitants, 6 = Urban unit of 100,000 to 199,999 inhabitants, 7 = Urban unit of 200,000 to 1,999,999 inhabitants, 8 = Paris urban unit.
- `tuu_r`: same with grouped values: 1 = Rural municipality, 2 = Urban unit of fewer than 20,000 inhabitants, 3 = Urban unit of 20,000 to fewer than 200,000 inhabitants, 4 = Urban unit of 200,000 or more inhabitants (excluding the Paris conurbation), 5 = Paris conurbation.

- `typvois`: type of housing or neighborhood: 1 = Scattered houses, outside built-up areas, 2 = Houses in housing estates, residential neighborhoods, or towns, 3 = Apartment buildings in towns (other than housing projects or large estates), 4 = Apartment buildings in housing projects or large estates, 5 = Mixed housing: both apartment buildings and houses.
- `occupation_logement`: occupancy status in the dwelling: 0 = Person permanently residing in the dwelling, 1 = Child in shared custody, 2 = Career military personnel housed elsewhere in barracks or camps, 3 = Boarding school pupil or student residing elsewhere in a communal facility (university residence or student hall), 4 = Pupil or student residing elsewhere in an independent dwelling, considered as occasional (excluding communal facilities) (the surveyed dwelling is that of the parents), 5 = Young person residing elsewhere in a young workers' hostel, 6 = Pupil or student housed in a dwelling other than their parents', considered as occasional (excluding communal facilities) (the surveyed dwelling is not the parents'), 7 = Person residing elsewhere in a sanatorium, hospital, care facility, or prison, 8 = Person residing elsewhere in a retirement home or hospice, 9 = Person (other than a student) residing elsewhere in an occasional dwelling (excluding communal facilities).
- `matri`: legal marital status: 1 = Single, 2 = Married or remarried, 3 = Widowed, 4 = Divorced.
- `nat`: nationality (fourteen values): 10 = French, 11 = Spain, 12 = Italy, 13 = Portugal, 14 = Other EU-28 countries, 15 = Other European countries, 21 = Algeria, 22 = Morocco, 23 = Tunisia, 24 = Other African countries, 31 = Turkey, 32 = Cambodia–Laos–Vietnam, 41 = Americas (American continent), 51 = Other countries (Asia or Oceania) or stateless, 99 = Foreign person of unspecified nationality.
- `acteu`: labor market status according to ILO definition (variable name for EU activity — harmonized definition at the European Union level): 1 = Employed, 2 = Unemployed, 3 = Inactive.
- `acteu_detail`: same with more detailed values: 1 = Employed, 3 = ILO unemployed actively seeking work (PSERE), 4 = Other ILO unemployed, 5 = Student, pupil, or trainee in education (inactive), 6 = Other inactive (including retired).
- `ancchom` and `ancinactif`: duration of unemployment and inactivity in months (see Stata label); note: 99 means 99 months or more.
- `halochomage`: halo around unemployment (partial characterization of ILO unemployed): 1 = Seeking work but not available, 2 = Available to take a job but not seeking one, 3 = Would like a job but neither seeking nor available.
- `csp_actif_detaille`: socio-professional category (SPC) for active individuals (detailed values): Empty = Not applicable (ACTEU other than 1 or 2), 00 = Not stated, 10 = Farmers, 11 = Small-scale farmers, 12 = Medium-scale farmers, 13 = Large-scale farmers, 21 = Craftsmen, 22 = Shopkeepers and related, 23 = Heads of companies with 10 or more employees, 31 = Liberal professions, 33 = Senior civil servants, 34 = Professors and scientific professions, 35 = Information, arts, and entertainment professions, 37 = Senior administrative and commercial managers in companies, 38 = Engineers and senior technical managers in companies, 42 = Primary school teachers and related, 43 = Intermediate health and social work professions, 44 = Clergy and religious, 45 = Intermediate administrative professions in the public sector, 46 = Intermediate administrative and commercial professions in companies, 47 = Technicians, 48 = Foremen and supervisors, 52 = Civil servants and public service workers, 53 = Police and military, 54 = Administrative employees in companies, 55 = Sales employees, 56 = Personal service workers, 62 = Skilled industrial workers, 63 = Skilled craft workers, 64 = Drivers, 65 = Skilled handling, warehousing, and transport workers, 67 = Unskilled industrial workers, 68 = Unskilled craft workers, 69 = Agricultural workers, 81 = Unemployed who have never worked.
- `csp_actif_int`: socio-professional category (SPC) for active individuals (intermediate values): Empty = Not applicable (ACTEU other than 1 or 2), 00 = Not stated, 10 = Farmers, 21 = Craftsmen, 22 = Shopkeepers and related, 23 = Heads of companies with 10 or more employees, 31 = Liberal professions, 32 = Senior civil servants, intellectual and artistic professions, 36 = Senior managers in companies, 41 = Intermediate professions in education, health, the public sector, and related, 46 = Intermediate administrative and commercial professions in companies, 47 = Technicians, 48 = Foremen and supervisors, 51 = Public sector employees, 54 = Administrative employees in companies, 55 = Sales employees, 56 = Personal service workers, 61 = Skilled workers, 66 = Unskilled workers, 69 = Agricultural workers, 81 = Unemployed who have never worked.

- `csp_actif`: socio-professional category (SPC) for active individuals (aggregated values): Empty = Not applicable (ACTEU other than 1 or 2), 0 = Not stated, 1 = Farmers, 2 = Craftsmen, shopkeepers, and company heads, 3 = Senior managers and higher intellectual professions, 4 = Intermediate professions, 5 = Clerical and service employees, 6 = Manual workers, 8 = Unemployed who have never worked.
- `contrat`: type of employment contract: Empty = Not applicable or not stated, 0 = No employment contract, 1 = Permanent contract (CDI), 2 = Fixed-term contract (CDD) other than seasonal, 3 = Seasonal contract, 4 = Temporary agency contract, 5 = Apprenticeship contract.
- `heuretra_tranche`: average number of hours worked per week in the main job (in brackets): 1 = Fewer than 15 hours, 2 = At least 15 but fewer than 30 hours, 3 = At least 30 but fewer than 35 hours, 4 = At least 35 but fewer than 40 hours, 5 = 40 hours or more.
- `typehoraire`: type of working hours in the main job: Empty = Not applicable (ACTEU other than 1) or not stated, 1 = Roughly the same from one week to the next, 2 = Alternating shifts: 2×8, 3×8, teams, etc., 3 = Variable from one week to the next, 4 = Not applicable (worked only one week...).
- `compenheuresup`: existence and, where applicable, nature of compensation for overtime hours: Empty = Not applicable or not stated, 1 = Yes, in the form of pay, 2 = Yes, in the form of compensatory rest (including recovery time), 3 = Yes, both in the form of pay and compensatory rest (including recovery time), 4 = No compensation, 9 = Don't know.
- `education`: highest qualification obtained (6 values): 1 = Degree above two years post-baccalaureate, 3 = Two-year post-baccalaureate degree, 4 = Baccalaureate or vocational baccalaureate or equivalent, 5 = CAP, BEP, or equivalent vocational qualification, 6 = Brevet des collèges (lower secondary school certificate), 7 = No qualification or primary school certificate.
- `education_detail1`: same (16 values): 10 = Master's (research or professional), DEA, DESS, Doctorate, 12 = Schools at licence level and above, 22 = Maîtrise (M1), 21 = Licence (L3), 30 = DEUG, 31 = DUT, BTS, 32 = Other qualification (bac+2 level), 33 = Paramedical and social (bac+2 level), 41 = General baccalaureate, 42 = Technological baccalaureate, 43 = Vocational baccalaureate, 44 = Technician's certificate or vocational diploma, 50 = CAP, BEP, 60 = Brevet des collèges, 70 = Primary school certificate, 71 = No qualification.
- `education_detail2`: same (11 values): 10 = Licence (L3), Maîtrise (M1), Master's (research or professional), DEA, DESS, Doctorate, 11 = Schools at licence level and above, 30 = DEUG, 31 = BTS, DUT or equivalent, 33 = Paramedical and social (bac+2 level), 41 = General baccalaureate, 42 = Technological baccalaureate, vocational baccalaureate or equivalent, 50 = CAP, BEP or equivalent, 60 = Brevet des collèges, 70 = Primary school certificate, 71 = No qualification.
- `csp_mere` and `csp_pere`: socio-professional category of the mother and father: 00 = Not stated, 10 = Farmers, 11 = Small-scale farmers, 12 = Medium-scale farmers, 13 = Large-scale farmers, 21 = Craftsmen, 22 = Shopkeepers and related, 23 = Heads of companies with 10 or more employees, 31 = Liberal professions, 33 = Senior civil servants, 34 = Professors and scientific professions, 35 = Information, arts, and entertainment professions, 37 = Senior administrative and commercial managers in companies, 38 = Engineers and senior technical managers in companies, 42 = Primary school teachers and related, 43 = Intermediate health and social work professions, 44 = Clergy and religious, 45 = Intermediate administrative professions in the public sector, 46 = Intermediate administrative and commercial professions in companies, 47 = Technicians, 48 = Foremen and supervisors, 52 = Civil servants and public service workers, 53 = Police and military, 54 = Administrative employees in companies, 55 = Sales employees, 56 = Personal service workers, 62 = Skilled industrial workers, 63 = Skilled craft workers, 64 = Drivers, 65 = Skilled handling, warehousing, and transport workers, 67 = Unskilled industrial workers, 68 = Unskilled craft workers, 69 = Agricultural workers, 71 = Former farmers, 72 = Former craftsmen, shopkeepers, and company heads, 74 = Former senior managers, 75 = Former intermediate professions, 77 = Former clerical and service employees, 78 = Former manual workers, 81 = Unemployed who have never worked, 82 = Other persons without professional activity.
- `nat_mere`, `nat_pere`, `origine`.: These three variables share the same values:
  - 01 = France

  - 03 = Northern Europe
  - 04 = Southern Europe
  - 05 = Eastern Europe
  - 06 = Maghreb
  - 07 = Rest of Africa
  - 08 = Middle East
  - 09 = Laos, Vietnam, Cambodia
  - 10 = Rest of the world
  - 99 = Not stated / Unknown country

  The first two are the nationalities at birth of the individual's parents. The variable origine is constructed as follows:

  - `origine` = 01 if both parents held French nationality at birth, i.e. `nat_mere` = 01 and `nat_pere` = 01. Otherwise, if both parents held the same nationality (in the sense of the values above) at birth, then origine equals that value: `origine` = x if `nat_mere` = x and `nat_pere` = x, for $x \in \{03, 04, 05, 06, 07, 08, 09, 10, 99\}$.
  - Or, if one parent held French nationality and the other held a different nationality, then origine equals the value of that second parent: `origine` = x `nat_mere` = x and `nat_pere` = 01, or if `nat_mere` = 01 and `nat_pere` = x, for $x \in \{03, 04, 05, 06, 07, 08, 09, 10, 99\}$.
  - Or, if both parents held non-French nationalities that differ from each other, then origine is set arbitrarily equal to `nat_mere` (using the father's nationality as the reference would only marginally change the results, as this last case is the rarest and least frequent in the data): `origine` = x if `nat_mere` = x and `nat_pere` = y, with $x \neq y$ and $x, y \in \{03, 04, 05, 06, 07, 08, 09, 10, 99\}$

  As needed in your analyses, certain values of the `origine` variable may be grouped together, or interactions with other variables may be considered, such as the `immi` variable, for instance.

- `sante_declaree`: individual's self-reported assessment of their health: Empty = Not applicable (intermediate re-interview, i.e. other than t = 1 or 6) or not stated, 1 = Very good, 2 = Good, 3 = Fair, 4 = Poor, 5 = Very poor, 8 = Refused, 9 = Don't know.